1 Clouds and Orographic Precipitation

Cloud microphysics in a single-column model

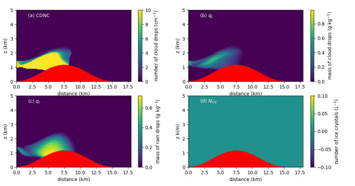
1.1 Changing cloud-droplet number

Prediction. Predict how increasing cloud-droplet number affects collision–coalescence when the amount of cloud water is similar.

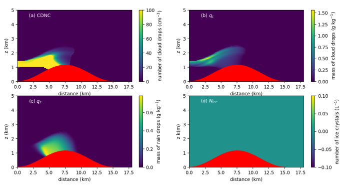
Expected prediction. Increasing droplet number divides the available liquid water among more droplets, so the typical droplet is smaller. Smaller droplets collide and coalesce less efficiently, so warm-rain formation should be delayed and weakened. Reducing droplet number should have the opposite effect.

Exercise 1. Where does rain develop in the default case, and how does changing droplet number affect the position and amount of rain?

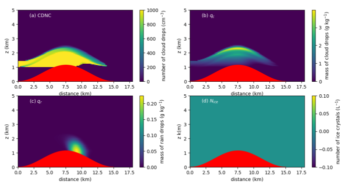
Model answer. In the default $N_d \simeq 100\ \mathrm{cm}^{-3}$ case, rain develops on the windward side and around the crest of the hill, then falls towards the surface. With only about $10\ \mathrm{cm}^{-3}$ droplets, the available cloud water is shared between fewer droplets, so individual drops grow larger and collision–coalescence is very efficient: rain forms earlier and the rain-water content is large. With $1000\ \mathrm{cm}^{-3}$, the droplets remain smaller, cloud water persists farther over the hill, and rain is much weaker and shifted downwind. The optional $2000\ \mathrm{cm}^{-3}$ case exaggerates this suppression still further.



(a) $N_d = 10\ \mathrm{cm}^{-3}$.



(b) $N_d = 100\ \mathrm{cm}^{-3}$.



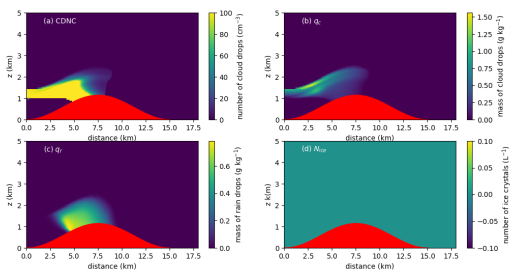
(c) $N_d = 1000\ \mathrm{cm}^{-3}$.

Figure 1: Increasing droplet number suppresses and delays warm-rain production while allowing more cloud water to persist.

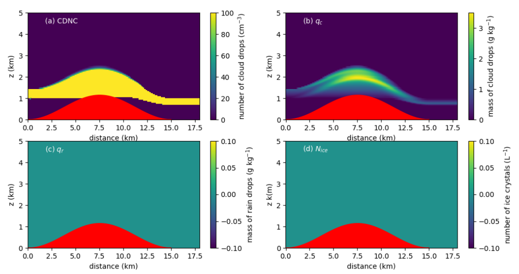
1.2 Switching off the warm-rain process

Exercise 2. What happens to cloud water and rain when collision–coalescence is disabled in the warm cloud?

Model answer. With `wr_flag=.false.`, the rain-water field disappears in the warm-cloud calculation and cloud water persists much farther over and beyond the hill. The model can still condense vapour into cloud droplets, but the principal pathway converting those droplets into raindrops has been removed. The contrast is especially strong at very high droplet number because collision–coalescence is already inefficient there; disabling it completely leaves a persistent cloud with essentially no warm rain.



(a) Warm rain on.



(b) Warm rain off.

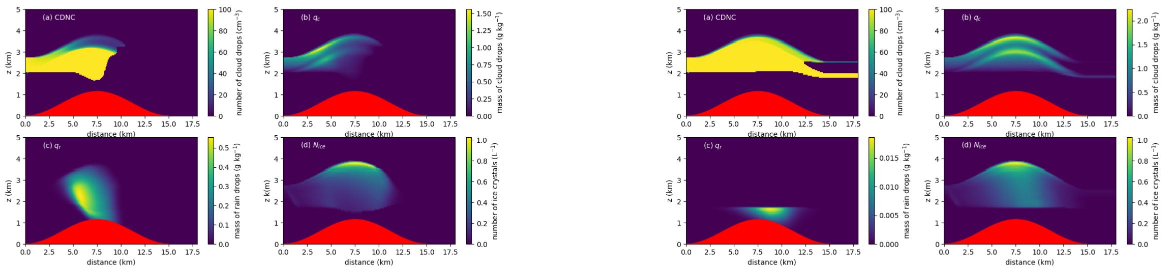
Figure 2: Controlled comparison at the default droplet number.

1.3 Cooler conditions: ice and precipitation pathways

For the supplied output, the cold-cloud experiment uses $T_{\text{surf}} = 290$ K, $T_{\text{cbase}} = 270$ K and $T_{\text{ctop}} = 265$ K.

Exercise 3. What happens when warm rain is switched off in the cold case? Is the response the same as in the warm cloud?

Model answer. The liquid rain-water field is strongly reduced when warm rain is disabled, but the cold cloud still contains a substantial ice-crystal population. This differs from the warm-cloud calculation, where there is no ice-phase route at all. The cold case therefore retains microphysical pathways involving ice even though liquid collision–coalescence has been removed. In the supplied fields the remaining liquid rain is small, but the non-zero ice field shows clearly that precipitation-forming microphysics has not been reduced to a purely warm-cloud problem.



(a) Cold cloud, warm rain on.

(b) Cold cloud, warm rain off.

Figure 3: The cold cloud retains a clear ice population even when the warm-rain pathway is disabled.

Optional coarse-aerosol / INP perturbation

The background aerosol population includes a small coarse second-mode concentration, $n_{\text{aer},2} = 0.1 \times 10^6$ kg^{-1}. The dust perturbation increases only this second number to 10×10^6 kg^{-1} while retaining the cold temperatures and `wr_flag=.true.`.

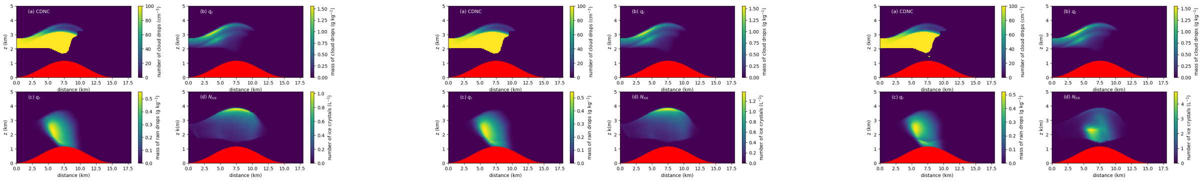
Exercise 4. How does the coarse-aerosol perturbation change ice-crystal number and precipitation relative to the cold reference?

Model answer. The coarse-mode perturbation produces a modest increase in ice-crystal number: the maximum plotted value rises from roughly 1 L^{-1} in the cold reference to around 1.2–1.3 L^{-1}. The cloud- and rain-water fields change much less than the ice-number field. In this SCM the primary-ice parameterisation depends on temperature and on the number of aerosol particles larger than 0.5 μm, so increasing the coarse population increases the diagnosed INP supply. This should be interpreted as a parameterised INP sensitivity, not as a complete mineral-dust chemistry calculation.

Optional secondary ice production with the dust mode retained

Exercise 5. What happens when the additional SIP mechanisms are enabled while the cold temperatures, warm-rain setting and dust mode are held fixed?

Model answer. Ice-crystal number increases much more strongly than in the primary-ice perturbation alone. In the supplied output the peak rises from about 1.2 L^{-1} in `scm_cold_dust.png` to roughly 4–5 L^{-1} in `scm_cold_dust_sip.png`. The broad cloud- and rain-water distributions remain recognisably similar, although there are local changes associated with the altered ice population. Secondary ice can amplify ice number because one primary ice particle can participate in processes that generate multiple additional ice crystals; the ice-number concentration is therefore not limited to the original number of activated INPs.



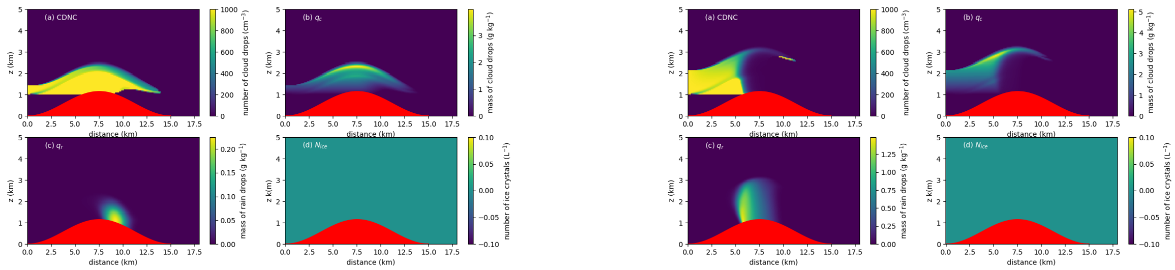
(a) Cold reference. **(b)** Coarse-aerosol perturbation. **(c)** Dust + additional SIP.

Figure 4: The coarse aerosol gives a modest primary-ice response, whereas the additional SIP switches produce a much larger increase in ice number.

1.4 Deeper cloud

Exercise 6. How does increasing cloud depth affect precipitation, and why?

Model answer. With $N_d = 1000\ \mathrm{cm^{-3}}$ held fixed, deepening the cloud from the shallow reference to the 5 K cloud-base-to-top temperature difference produces much more cloud water and substantially more rain. Rain also develops earlier, nearer the windward/crest region, rather than being confined to a weak downwind pocket. A deeper cloud gives droplets more vertical distance and more time in cloud to grow by condensation and collection. This increases the probability that some droplets reach sizes for which collision–coalescence becomes efficient, overcoming much of the suppression associated with the high droplet number.



(a) Shallow cloud, $N_d = 1000\ \mathrm{cm^{-3}}$. **(b)** Deeper cloud, same N_d.

Figure 5: Increasing cloud depth strongly enhances precipitation even though droplet number is unchanged.